EXPERIMENT - 4

Aim : To develop a blockchain-powered web application using the Solidity programming language on Remix IDE and integrate it with the MetaMask wallet.

Theory

Blockchain

Blockchain is a distributed digital ledger that stores transactions in blocks. These blocks are linked together using cryptographic techniques. It provides transparency, security, and immutability of recorded data

Solidity

Solidity is a high-level programming language used to develop smart contracts on Ethereum-compatible blockchain networks. It allows developers to define rules and operations that execute on the blockchain.

Remix IDE

Remix IDE is a web-based development environment used to write, compile, deploy, and test Solidity smart contracts. It provides tools to interact with smart contracts without requiring a complete local development setup.

MetaMask Wallet

MetaMask is a cryptocurrency wallet that allows users to manage blockchain accounts and connect decentralized applications (DApps) to blockchain networks. It enables users to approve transactions and interact with smart contracts.

Testnet

A testnet is a blockchain network used for development and testing. It allows developers to deploy and test smart contracts using test tokens instead of real cryptocurrency. Sepolia is an Ethereum testnet.

Smart Contract

A smart contract is a program stored and executed on a blockchain. It automatically performs predefined operations when the required conditions are met. Smart contracts help reduce manual intervention and improve transparency.

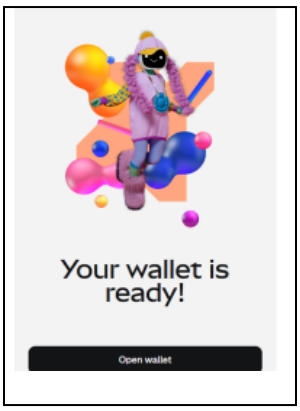
Connecting MetaMask with Remix IDE

MetaMask can be connected to Remix IDE using the browser wallet provider. After selecting the required test network in MetaMask, the user selects the corresponding injected provider in Remix. The user can then deploy smart contracts and perform transactions after approving them through MetaMask.

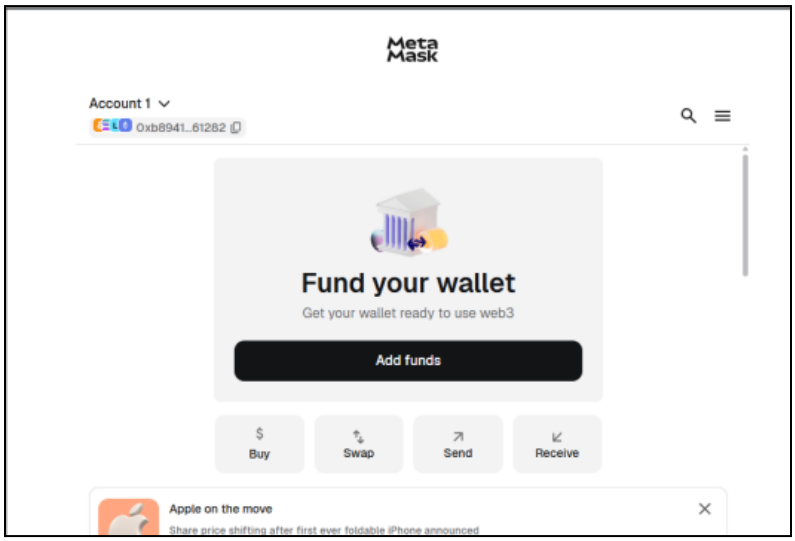
Contract Deployment and Transactions

Contract deployment stores the compiled smart contract on the blockchain. Transactions such as updating data or calling functions that change blockchain state require approval and consume gas. Transaction details can be inspected using a blockchain explorer when the selected network is supported by that explorer.

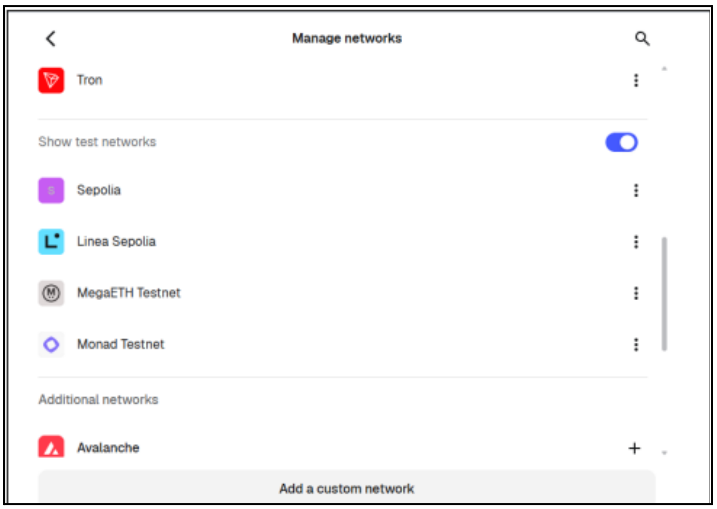
Implementation



MetaMask Wallet Successfully Created and Ready to Use



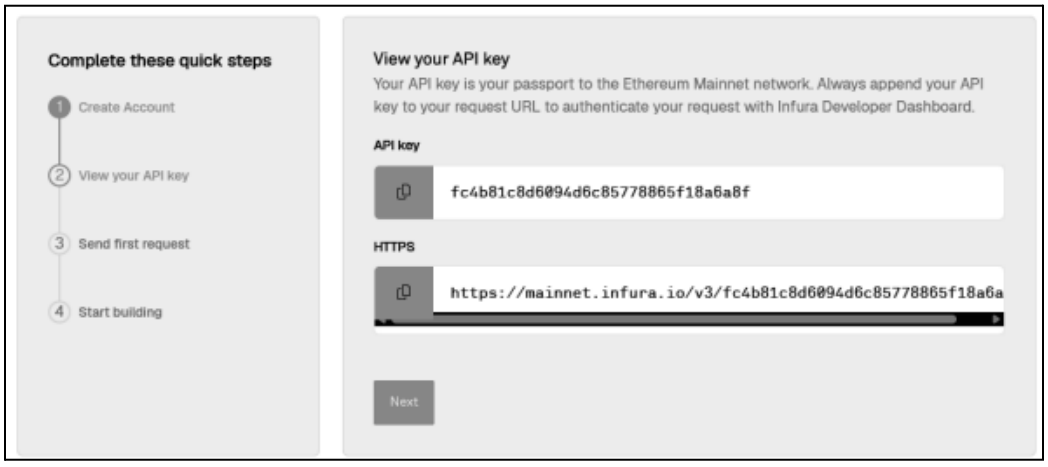
MetaMask Wallet Dashboard Showing Account and Funding Options



MetaMask Interface Showing Sepolia Test Network



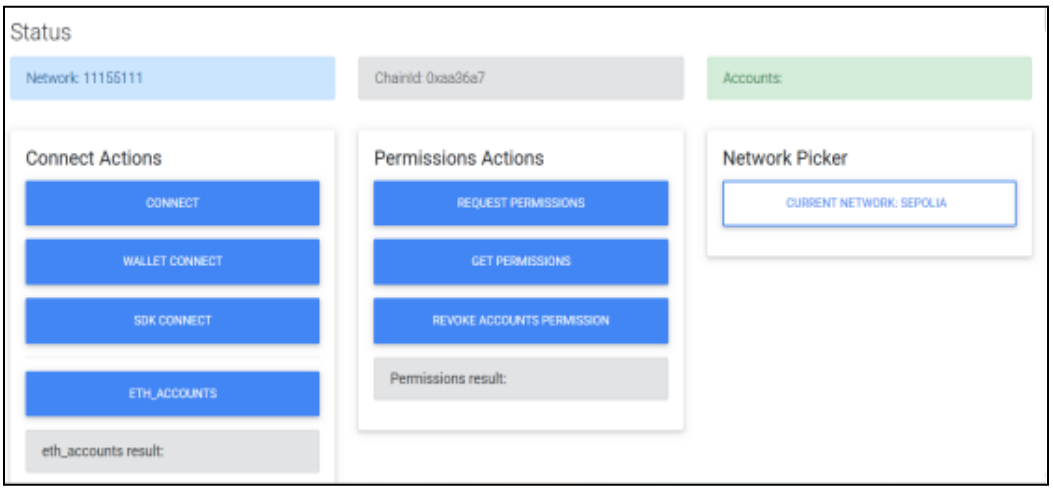
MetaMask Wallet Connected to Remix IDE



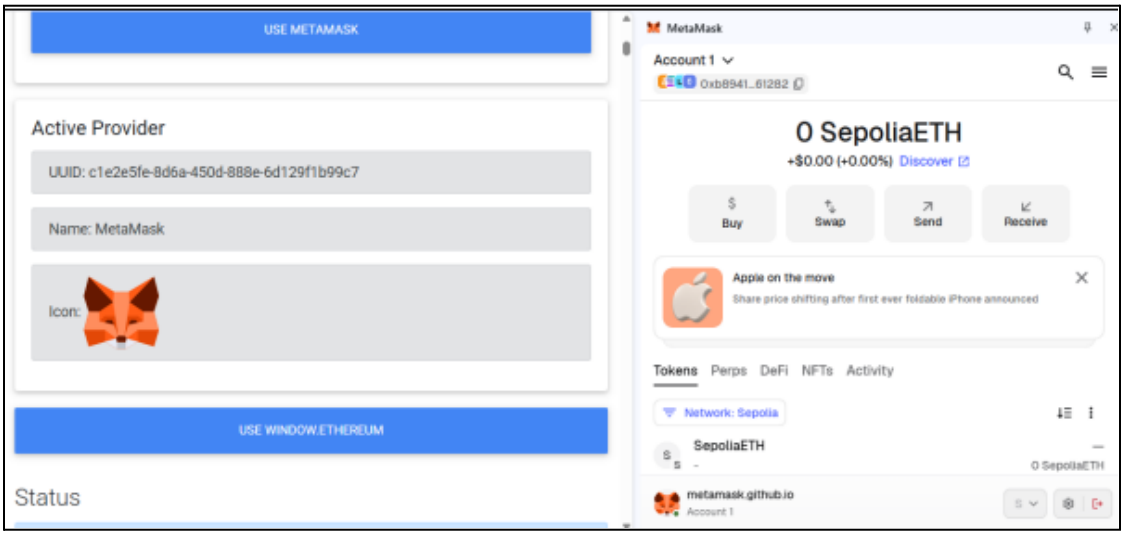
Infura Dashboard Displaying the Generated API Key and HTTPS Endpoint



Infura Dashboard Showing the "Send Your First Request" Step



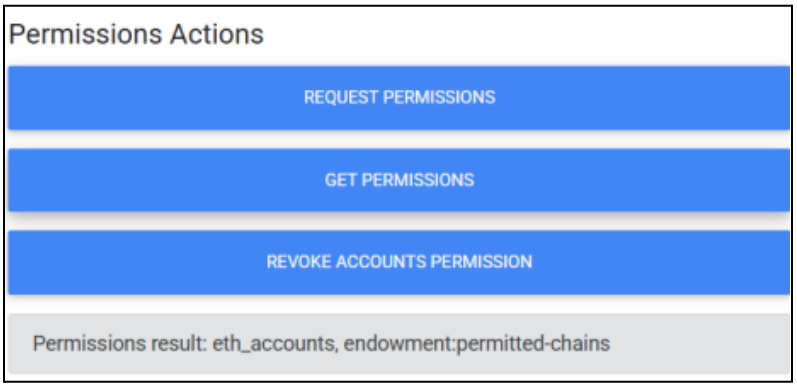
MetaMask Test DApp Showing Sepolia Network and Connection Actions



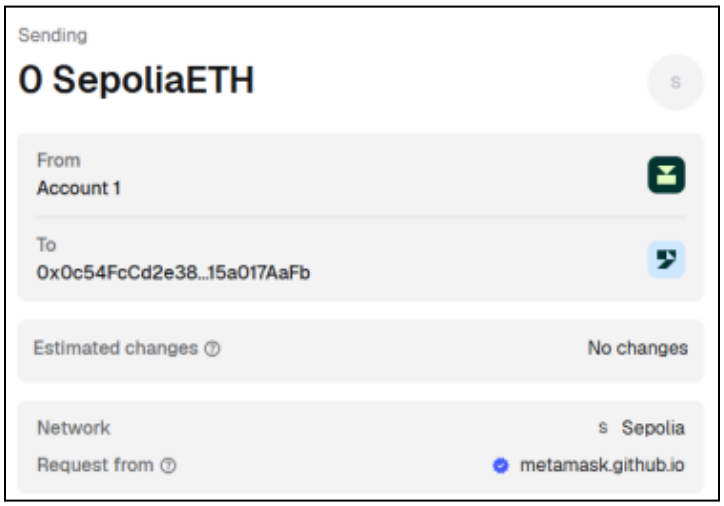
MetaMask Test DApp Showing Active MetaMask Provider and Sepolia Wallet Connection



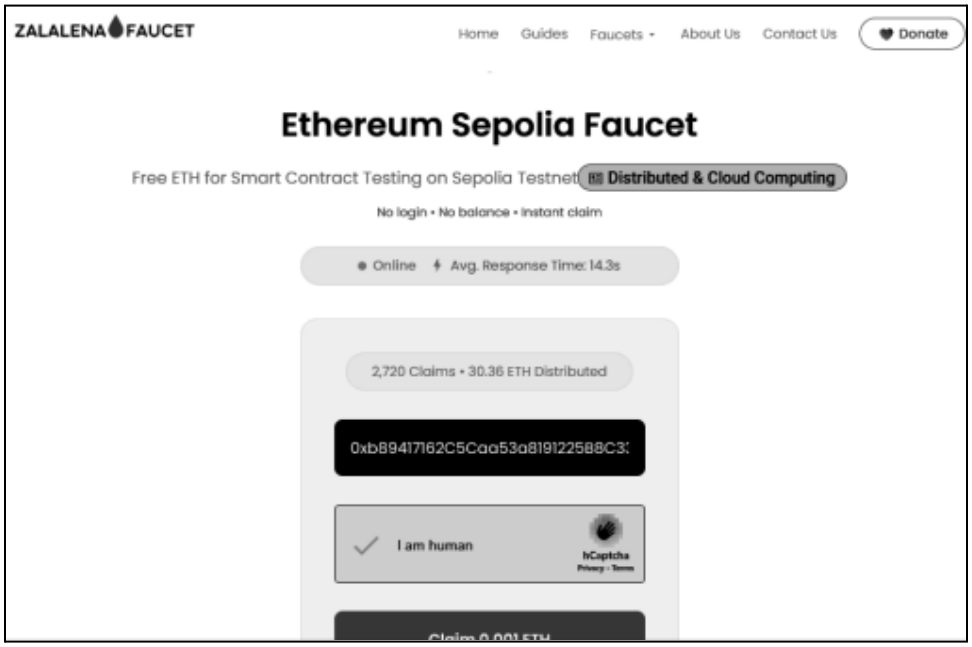
MetaMask Test DApp Successfully Connected and Wallet Address Retrieved Using *eth_accounts*



DApp Permissions Result Showing Authorized Account and Permitted Chains



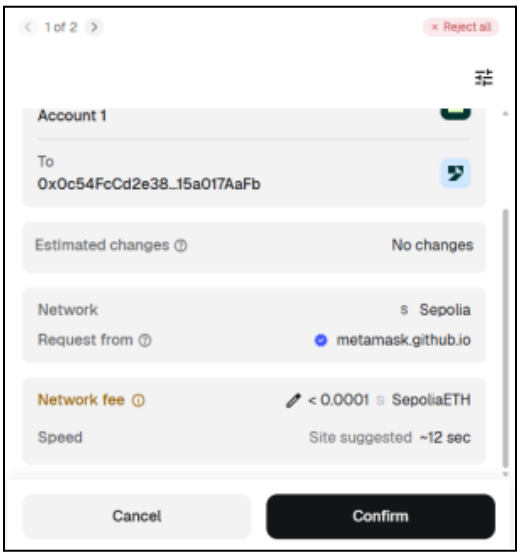
MetaMask Transaction Confirmation Request on the Sepolia Test Network



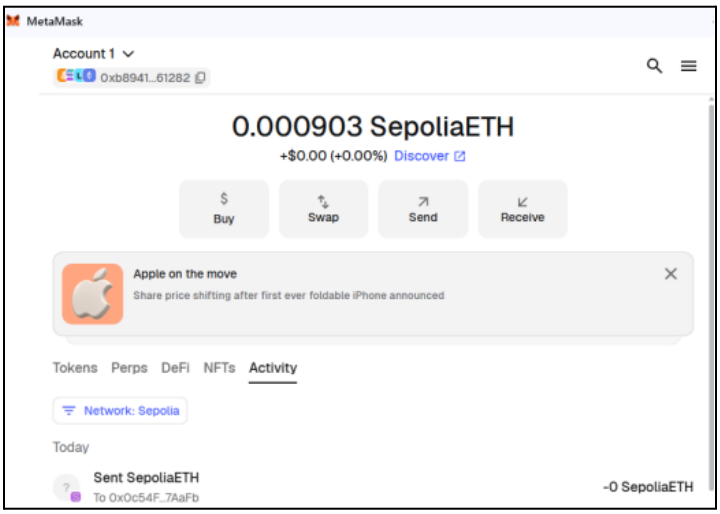
Requesting Sepolia Test ETH from a Cryptocurrency Faucet



Successful Sepolia Test ETH Claim from Zalalena Faucet



MetaMask Transaction Confirmation Screen for an EIP-1559 Transaction on Sepolia



MetaMask Activity Tab Showing Successful Sepolia ETH Transaction

Conclusion

The experiment demonstrates the development and deployment of a blockchain-powered web application using Solidity and Remix IDE. MetaMask was used to connect a wallet account to the blockchain and approve transactions. The experiment helped understand smart contract deployment and interaction on a test network.